UM Hackathon 2026

Malaysia LRT

LRT AI Operations

Reduce Cost · Improve Productivity · Support Staff Decisions

Kelana Jaya Line

Ampang Line

Sri Petaling Line

18:00–19:00 | 20 trains/hr

Expected load | 87%

GLM Rec | MODERATE ✓

Event: Blackpink Concert

Weather: Rainy +20% pax

Extra cost | RM 2,100

POWERED BY Z.AI GLM-5.1

THE CHALLENGE

Scheduling LRT trains is harder than it looks

3

LRT lines to manage

Kelana Jaya, Ampang, Sri Petaling - each line runs 18 hours a day with different passenger patterns on every route.

4

variables every decision touches

Weather, events,andemergenciescanallhit at once - and each one changes how many trains you actually need.

Events bring unexpected crowds

A concert or football match floods stations with thousands of extra passengers - the standard timetable wasn't built for that.

Emergencies need fast calls

When something goes wrong - track incident, power failure, signal fault - staff have minutes to decide with no clear playbook.

OUR SOLUTION

LRT AI closes the gap in real time

A tool built for LRT staff - pick the day, describe what's happening, and the system tells you exactly how many trains to run and why.

Pick a day, see the full schedule

Choosea line and date - get the full 18-hour timetable with trains, load, and cost shown per hour.

Enter the situation, get 3 options

Tell the system what's happening - weather, event, or emergency - and it instantly returns three options to compare.

AI picks the best one and explains why

GLM-5.1 recommends the best option with a clear reason. Staff confirm, apply it to the schedule, and save - done.

Update and save the schedule

One click applies the chosen option to the full-day timetable - saved to the database and ready for any shift to retrieve.

RM 350

cost per train-hour → every decision shows exact cost impact

75%

target load factor → optimise trains without overspending

3 factors → 3 options → 1 pick

Weather · Event · Emergency - GLM-5.1 weighs all three and recommends

HOW IT WORKS

Four steps from problem to updated schedule

1 INPUT

Tell the system what's happening

Staff enter the weather, any nearby events, or an emergency - via a form or plain text.

Weather | Event | Emergency

→

2 COMPUTE

System calculates 3 options instantly

A math engine works out three train frequency options - with load, passengers, and cost shown for each.

Conservative | Moderate | Aggressive

→

3 RECOMMEND

AI picks one and explains

GLM-5.1 picks the best option and gives staff a clear reason why - in plain language, not technical jargon.

Live streaming | Plain language

→

4 IMPLEMENT

Apply and save

Staff pick an option, apply it to the full-day schedule, and save it to the database - ready for any shift to load anytime.

One click apply | Saved to DB

SITUATION INPUT

Two ways to describe any situation

01

Manual Inputs

Pick weather, event type, crowd size, and emergency type from simple dropdowns - no typing needed.

Weather: **clear / cloudy / rainy / stormy**

Event: **concert, football, festival...**

Emergency: **6 incident types**

Rain caps max freq + boosts event ridership

Instant - no API call

02

Free-Text Description

Just type what's happening in plain English - GLM-5.1 reads it and figures out the details automatically.

"Blackpink concert at Bukit Jalil,
~4,000 extra pax/hr. Rainy."

GLM extraction

RECOMMENDATION ENGINE

The system does the hard thinking for you

CONSERVATIVE

−2

TRAINS/HR VS MODERATE

Fewer trains, lower cost. But trains get crowded - some passengers may be left waiting on the platform.

Risk: overcrowding

MODERATE

✦ GLM Pick

Optimal

DEMAND-MATCHED FREQUENCY

The right number of trains for the situation - passengers are comfortable and cost stays in check.

Best balance

AGGRESSIVE

+3

TRAINS/HR VS MODERATE

Extra trains for extra safety. Costs more - but worth it when demand is hard to predict.

High uncertainty

GLM-5.1 REASONING ▶ LIVE

"The primary factor is the Blackpink concert at Bukit Jalil - 4,000 extra pax/hr, compounded by rainy weather adding ~20% ridership shift from cars to LRT...

Moderate at 17 trains/hr achieves 81% load - within comfort range. Conservative risks 97% load with queue spillover. Aggressive adds RM 1,050/hr beyond what demand justifies..."

RECOMMENDATION: moderate

Staff can always **override** the AI pick and go with a different option if they know better.

SAFETY-FIRST DESIGN

Every emergency has a clear response

TRACK

Track Incident

Someone on the track - all trains stop immediately, no exceptions. After clearance, max trains deployed to clear the backlog.

FULL SUSPENSION

SIG

Signal Failure

Signals protect trains from collisions. If they fail, everything stops until the fault is fixed - no exceptions.

SERVICE HALT

PWR

Power Failure

Service drops to backup power - fewer trains running, loads get higher. System shows what's still safe to operate.

REDUCED SERVICE

BRK

Train Breakdown

One train is out - the rest keep going. Just slightly fewer per hour until it's back in service.

MINOR REDUCTION

EVAC

Evacuation

Get everyone out fast. Maximum trains, no debate - safety comes first, always.

MAX FREQUENCY

OVR

Overcrowding

Platforms dangerously packed. System flags it right away and recommends more trains to clear the crowd quickly.

URGENT ADD

UNDER THE HOOD

Technology stack

Python

Core logic

Handles all the math - passenger calculations, option scoring, and saving schedules to the database.

Streamlit

UI framework

The interface staff actually use - clean dashboard, live schedule table, and AI reasoning streamed in real time.

GLM-5.1

Z.AI LLM

The brain of the system - reads the situation, picks the best option, explains why, and writes the shift briefing live.

SQLite

Schedule persistence

Every saved schedule stays in the database - staff on the next shift can pull it up anytime, no need to redo it.

pandas

Schedule table

Colour-coded rows: red emergencies, green events, blue adjusted.

INPUT LAYER

Manual · Text · Image

↓

MATH ENGINE

3 options - instant

↓

GLM LAYER

pick + reasoning

↓

UI LAYER

schedule · briefing

ROADMAP

What's next

01 **Live RAPID data integration**
Hook into real gate count data from RAPID - so passenger numbers are actual, not estimated.

High priority

02 **Multi-line coordination**
When Masjid Jamek gets hit by a big crowd, all three lines should adjust together - not just one.

High priority

03 **Push alerts to operations staff**
When the schedule changes, station staff get notified automatically - no need to call around manually.

Medium priority

04 **Outcome feedback loop**
Log what actually happened after each event - so the AI gets smarter and more accurate over time.

Medium priority

LIVE DEMO

Demo

Walkthrough ofthe LRT AI Operations system - live scheduling, situation input, and AI recommendation in action.

Schedule view | Situation input | AI recommendation | Save to DB